Public Expressions of Metrics in News Media: A Computational Textual Analysis

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Introduction

This paper serves as an exploratory analysis of how news media outlets communicate with their audiences about the use of metrics within their industry. Considerable work has been done examining how user engagement data has been integrated, challenged, and synthesized into the media workplace and between writers, but how these changes are presented and communicated to audiences remains largely unexplored. Using topic modeling to explore the semantic ties found within metrics focused news articles and publicly released strategy memos from a wide diversity of outlets, some common trends and subjects in audience facing discussion of metrics are explicated.

In the subsequent textual analysis used to explore these nuances, the following hypotheses will be used as guideposts for better navigating through the data, though germane and interesting aspects of each exploration will also be given their due.

Hypotheses

H1: Print outlets will publish less about metrics

H1 predicts that, since the majority of traditionally print outlets receive their funding and influence from their physical papers, print outlets will discuss metrics less in an audience facing manner.

H2: Financial topics will be a small minority of the text, except for business and strategy documents

H2 predicts that, where discussion of metrics occur in non-business and strategy documents, financial topics will be less prevelant than nearly every other topic. This is not due to journalists' ignorance or indifference towards the financial implications of metrics, but instead due to the professional cognitive dissonance between journalist's detached, objective ideals and the implications of having to modify their work to meet financial goals and needs [1].

H3: Targeted and reflective documents will be descriptive, untargeted will be prescriptive

H3 predicts that documents referencing a specific outlet, whether the publisher of the document or other outlet(s), will contain less normative and position-taking language as a product of journalistic norms. Instead of positing that outlets should utilize metrics in a specific manner, much of the targeted and reflective documents within the corpus will use their targeting or reflection as a tool to distance or relate themselves to others.

Topic Modeling

Topic modeling is a common form of statistical modeling used to discover the "topics" distributed across a number of documents; these topics are simply clusters of related words, most commonly

[1] Vu, H. T. (2014). The online audience as gatekeeper: The influence of reader metrics on news editorial selection. Journalism, 15(8), 1094–1110. https://doi.org/10.1177/1464884913504259

	Topic 1	Topic 2	Topic 3
Document 1	0.00	0.00	0.99
Document 2	0.00	0.99	0.00
Document 3	0.99	0.00	0.00

	Topic 1	Topic 2	Topic 3
Dog	0.00	0.00	0.99
Cat	0.00	0.99	0.00
Snake	0.99	0.00	0.00

identified through repeated proximity. Topics range from the obvious, such as a collection of mechanical terms extracted from a corpus of automotive magazines, to the more latent, such as emotionally charged language being identified in political speeches. As a tool, it is most frequently used to identify the latter form of topics.

The earliest work on topic modeling is traced to Thomas Hofmann, though this memo draws on a derivative of Hofmann's Probabilistic Latent Semantic Indexing[2] model known as latent Dirichlet allocation[3], or LDA. LDA is essentially two matrices of probabilities: the first is the probability of selecting a particular topic from a particular document, while the second is the probability of selecting a particular word (or phrase) from a particular topic

Above are simplistic examples of the two aforementioned matrices that could be produced by LDA. The leftmost table represents three topics and their probabilities of containing the word dog, cat, or snake, and the rightmost table shows how often each topic is found in a number of documents. By cross referencing the tables, one could infer that the first two documents are heavily related to cats and dogs, respectively, while the third could be a tragic tale of a chihuahua's run-in with a python.

These inferences, however, signify one of topic modeling's inherent weaknesses. While LDA is efficient and effective at mapping semantic valences and connections difficult to derive without computational aid, interpretation becomes difficult without the passive knowledge derived from manually parsing and coding these connections. The previous example is, again, simplistic: each topic is comprised of a single word, each of which appear cleanly in only a few documents. However, without knowing the context of these documents, only make surface level inferences about their nature can be made. The first assumption made of cats and dogs can be viewed as accurate, but only because the amount being inferred is modest. The second assumption could be correct, but without further detail of the topic or experience with the document is spurious.

Methodology

Data Source and Wrangling

Using an automated webcrawler, 62 digitally published documents were scraped from a sample of 68 news media websites[4]. To pull relevant documents, each crawler was instructed to parse through a given domain and search for metrics and analytics related keywords in both the URL and the body text, in this case defined by

HTML tags, of each page it found. From this subset, any pages containing a different set of COVID and sports related keywords in the URL or body were automatically discarded to make manual coding and sorting easier.

The result of the webcrawler was a scraped corpus of HTML documents discussing metrics in

[2]Hofmann, T. (1999, August). Probabilistic latent semantic indexing. In Proceedings of the 22nd annual international ACM SIGIR conference on Research and development in information retrieval (pp. 50-57).

[3]Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. Journal of machine Learning research, 3(Jan), 993-1022.

[4]See Appendix B for a complete list paired with the number of documents from each site

some form. The 189 document corpus was then manually filtered to remove documents unable to be manually filtered; finance reporting, for instance, often uses metrics language to describe market trends or financial news, but given the possibility that documents relevant to this work may discuss the impact of metrics on newsroom financial success, manual sorting was opted for.

Data Coding and Subsetting

The overall corpus is then split into six categories to better explore the proposed hypotheses and to give meaningful structures to the data.

The first two categories are Digital publications and Print publications; the distinction between these is whether the outlet began life as a print newspaper or as an online-only medium. The rationale behind this differentiation is to account for whether metrics, often digital in nature, are of equal concern to print publications as to digital publications. Print papers who reach far beyond their locale (NYT, Washington Post) have been labelled as digital for the purposes of this work. All documents in the corpus fit into one of these categories.

The next four distinctions have to do with content. Business and Strategy documents were marked as such to indicate those documents that did not fit the traditional news publication format; they instead serve as audience facing announcements, explanations, and debriefings of why and how an outlet is integrating, or avoiding, metrics into their workflow. These documents serve a notably different purpose than the average news media publication, and are thus marked, but not removed fom the Digital and Print pools.

The final three categories which are also coded for but not separated from the Digital and Print subsets deal with the targeting of the document. In writing about metrics, many documents referenced their own outlet or other outlets to typify the abstract discussion of traffic and numbers. In order to suss out any potential differences between documents that spoke about their (the author's or outlet's) use of metrics and others' use of metrics, these two categories were coded. In addition, a differentiation is made between those documents that target another outlet in name and those that speak broadly about the industry without specific mention; the resulting coding categories are Targeted and Untargeted.

Data Cleaning

The first and most important step in prepping an LDA model for this application is prepping the input. Natural language data is messy and full of noise in the form of stop words. Stop word is a catchall name for prospective terms that add little to no meaning to possible topics, namely function words such as the, is, at, which, and on. A further issue is punctuation, spacing characters and letter case; when converting text to data, a model can interpret these as separate terms.

In addition to the removal of stop words, those terms which will be fed into the model are first lemmatized. To lemmatize a word is to reduce it to its uninflected form, known as a lemma. For instance, the present tense verb discusses and the gerund discussing would both be reduced to the core lemma, discuss. This reduces noise by eliminating semantic redundancy in the topics and allows for more precise modeling of the the subjects being discussed.

Coherence and K-Topic Assumptions

Two additional aspects of LDA which are important for interpretation and model construction are the coherence of each topic and the number of topics that the model create.

Topic coherence is a measure of the semantic similarity between the most common words within each topic[5]. A topic coherence score closer to 0 indicates that the topics are more likely to be contaminated with artifacts of the statistical inference used by the model, and as the score grows closer to 1 the most common words for each topic grow closer in similarity.

For this work, topic coherence is taking a secondary role in evaluating models for two reasons. The first is that topic coherence is understood as interpretability measured by pairwise word-similarity scores. The models in this paper, however, are being used to understand conversational relationships between disparate subjects, not classifying and predicting discrete categories of terms.

The second reason for brushing aside topic coherence scores is the nature of the data being worked with. Natural language discussions are commonly winding and liable to touch on a number of subjects and interject these conversations with a variety of perspectives and non-sequiturs. Such data muddies otherwise cleanly categorized topics and thus hampers the coherence score, despite the fact this work relies on the tangential statements, varying language and perspectives.

Given these two concerns, coherence scores will still be computed, but only used as indicators of the general functioning of the models. In this role, an arbitrary range of $.2 < x < .5$ will be used to determine functionality, with upper and lower bounds chosen to eliminate models completely based on inferential artifacts as well as models with no inter-subject associations.

Coherence scores are tied with the number of topics a model is instructed to suss out. Choosing too few topics will cluster many disparate terms into a few options, while choosing too many topics will artificially increase the coherence score by producing arbitrary distinctions between many small topics, thus inhibiting interpretability of the overall model. After cleaning the data for each model, coherence scores for a number of possible topics were run for each model. A topic number of 8 is found to be within the aforementioned coherence score range for each model, thus each model is instructed to find 8 topics for consistency.

Coherence score charts for each model can be found in Appendix A

Interpreting Results

As there is a touch of subjective interpretation needed to contextualize and understand the product of topic modeling, I will be dedicating a subsection to expanding on possible interpretations and rationals behind each model's output, as well as their implications. Attached in a footnote to each subsequent LDA analysis will be a link to an interactive visualization of both salience and intertopic distance, calculated through multidimensional scaling. Under each location subsection, support or opposition towards the aforementioned hypotheses will be explicated in terms of topic content. The visualizations are interactive so that terms and topics mentioned in the analyses can be highlighted and personally examined. These visualizations have been removed from this paper for space; it is recommended you reference them alongside the analyses.

The leftmost graph employs multidimensional scaling to visualize the level of similarity between topics, as well as the prevelance of each topic within the corpus. In this example, the Jensen-Shannon divergence[6] between the probability distributions of each topic is mapped onto a plane

[5]Newman, D., Lau, J. H., Grieser, K., & Baldwin, T. (2010, June). Automatic evaluation of topic coherence. In Human language technologies: The 2010 annual conference of the North American chapter of the association for computational linguistics (pp. 100-108)

[6]Hinrich Schütze; Christopher D. Manning (1999). Foundations of Statistical Natural Language Processing. Cambridge, Mass: MIT Press. p. 304. ISBN 978-0-262-13360-9.

The rightmost chart contains term salience[7] information. It's default state shows the most salient terms for the entire corpus, but upon highlighting a topic on the Intertopic Distance Map the Salience Chart highlights the most salient terms for the chosen topic in addition to the corpus-wide frequency of the term.

Corpus Distribution

```
##                Digital Print Biz & Strat Targeted Reflective Untargeted
## # of Documents      46    16           9       22          5         26
```

The above table shows how the results of the webscrape, manual sorting and coding fell. As a reminder, Digital and Print publications constitute the entirety of the corpus, as do Business and Strategy, Targeted, Reflective and Untargeted, all four of which are distributed amongst the Digital and Print categories.

There is a notable gap between how many documents self reference and how many reference other outlets. This could be due to authors' unwillingness or discomfort to analyze their own work and environment in such a public manner, though it could also be attributed to restrictive company policy, filtering from editors, or possible ignorance of their outlets exact processes. Regardless, the number of reflective documents is small enough to render the LDA model uninterpretable. A case study will be included to demonstrate an example, cut topic modeling will be exlcuded.

Digital LDA[8]

From the most salient terms for this subset of the corpus, it's apparent that much of the discussion revolves around the impact of medium and platform. Terms such as (video, medium, facebook tv, live, platform, internet) are found within the most salient terms, indicating common references to audiovisual and social media forms of news. Specifically, Topic 6 contains references to (video, watch, live, facebook, view, minute), indicating that a number of documents reference the booming phenomenon of FaceBook video engagement. The intertopic distance of Topic 6 from other topics indicate that these subjects often stand alone within their own documents.

Another particularly interesting cluster is Topic 3, indicating a handful of documents made reference to abstract concepts such as (value, ethic) and even reference authorities within the field such as (poynter). This could hint at a possible integration of abstract journalistic codes and values into audience facing metrics discussions, as well as possible deference to authorities within the field. However, Topic 3 is distanced from the rest of the topics, indicating that these subjects are rarely integrated into other discussion topics.

Lastly, Topic 4 contains (chartbeat, company, editor, culture), hinting that some of the documents discuss a more practical, logistics oriented version of metrics. Environmental factors surrounding the journalist, such as the company at large, editorship, metrics tools and surrounding culture, could

Gawker Case Study: Internet Video Views Is A 100 Percent Bullshit Metric

Gawkers' 2016 piece contains some sobering self awareness:

[7]Röder, Michael, Andreas Both, and Alexander Hinneburg. "Exploring the space of topic coherence measures." Proceedings of the eighth ACM international conference on Web search and data mining. 2015.

[8]https://lukedanielcross.github.io/files/digitallda.html

> "But just as many newspapers have died and even the most powerful are still struggling to evolve—hey, look, in the past three months the New York Times added digital subscribers but still lost a bunch of money—the emphasis on video is going to produce way more losers than winners. The first losers are digital journalists paid to write, not make video (hi!), but soon many digital media companies that have bet heavily on video will be losers too."

This document, and this particular quote, illustrate an influential undercurrent in the digital documents of this corpus: digital journalists are starkly aware of metrics, particularly metrics within those social media and platforms that occupy a symbiotic relationship with their work, but they rarely engage with metrics on their home turf. Instead, discussion of metrics and journalism is delivered in a wrapper of FaceBook manipulation, or platformatic differences:

> "Facebook is nobody's friend, and especially not journalists'. They want to make live video look as attractive as possible to suck news organizations into providing high-quality content that users will engage with more than Grandpa Edwin's latest rant about Benghazi. They are giving live video priority in your feed and push notifications about videos are sent to followers, all of which serve to get more eyeballs onto videos"

This is a trend common among the digital outlet documents, and why FaceBook is such a salient term in the LDA. There is clearly an awareness of analytics influencing journalism, but that topic is never addressed on its face.

> "It is worth noting that the most successful Facebook Live videos so far have essentially been gimmicks; the type of things more palatably referred to as "content" than "journalism." And that, right there, is likely to be the future of digital media, and increasingly its present too. Multimedia production shops that very much resemble TV studios—and make no pretense at having a wall between reporting and advertising—thriving, while more traditional journalism shops attempt to eke out a hardscrabble existence."

This graf is the closest this document comes to discussing how journalism is being shaped by metrics. The rest of the text is dedicated to the influence of FaceBook on journalism, not how journalism reacts or adapts. Instead of openly discussing, and potentially evaluating, how journalistic processes are formed by metrics, many documents in the corpus use platform, format, or exogenous influences as a proxy for discussing the very metrics occuring within their own outlets.

Print LDA[9]

The most salient terms for print outlet documents indicate a greater interest in how metrics impact newspapers and the newsroom in particular. Instead of shifting focus to editors and platforms, more reporting positions such as (reporter, writer) are more commonly mentioned, as are traditionally journalistic terms such as (newsroom, reader). This doesn't necessarily indicate that print documents are directly addressing metrics and how they change journalism, however; many of these terms appear to be contextualized with temporal terms and terms of change (new, old, change, build, year). These terms hint at a focus on the shifting times, and how platforms and journalistic evolution interact with metrics.

[9] https://lukedanielcross.github.io/files/printlda.html

One anomaly from the print corpus subset is the focus on hierarchy and pay. Topic 4 hosts an intermingling of terms (writer, measure, pay, value, volunteer) which could hint metrics discussions being tinted with salary and measuring success in print documents.

Atlanta Journal-Constitution Case Study: Learning to adapt in a digital age

As demonstrated by the LDA, print documents focus heavily on an outlets change and evolution:

> "Like most newspapers across the country, The Atlanta Journal-Constitution is focusing on becoming an audience-first (or digital first) newsroom. Instead of merely trying to get people to come to us, we're now going after the audience. We recently reorganized our newsroom in topic teams that will allow reporters (who are closest to readers) and their editors to have more say about what they cover and how they cover it. We try to get our stories online faster and use research and metrics to inform us more about what readers want."

In contrast to the pervasive negativity found in the digital outlet documents, print outlet documents have an air of hopefulness and positivity:

> "One student said the only time she thinks to buy a newspaper is when there's "historic" news. Then, she'll buy two papers – one to read and one to save. For an editor who's a leader for the print edition, that can be a hard pill to swallow. Luckily, they didn't tell me anything I hadn't already heard. More than anything, it validated where we're headed as a company and as an industry."

This could be attributed to the perceived direction of evolution; all the print newspapers sampled were chosen for their size and success, in hopes of finding more data. With an already established base readership and business model for their print editions, expansion into digital metrics is seen as a safe move upwards instead of a forced move to survive.

> "It's my hope that in the coming year, that our newsroom will not only learn more about Millennials and Digital Natives, but that we'll start to figure out a good way to include them more in our newspaper products – give them something they'll want to read and share with their peers."

Business and Strategy LDA[10]

The most unique aspect of the business and strategy category is a reliance of marketing terms (marketing, marketer, advertisement, brandvoice). Topic 1 is entirely dedicated to this area, and occupies over a third of all terms.

Topic 4 is unique for how it summarizes teams and working groups within an outlet (team, coverage, help, join). This exemplifies a concept which shows up in a later NYT case study, showing the use of metrics is discussed in terms of diffusion, with a number of roles playing a different part in the metrics implementation process.

[10] https://lukedanielcross.github.io/files/businesslda.html

Targeted LDA[11]

The targeted category falls largely in line with the digital category, with a few noteworthy exceptions. Topic 1 showcases the trend where, in examining another outlets approach to metrics, the documents seemingly employ a higher level of analysis than the gore and detail found in digital documents. Terms including (culture, model, platform) indicate an more distanced form of conceptualizing how these metrics models function.

Similarly, (audience, reader, people) sit higher in the salient terms list, indicating these topics are mentioned more often. This could be attributed to H3 and the division of descriptive and prescriptive language, as documents in this category are more likely to describe audience interactions in other outlets rather than prescribe a change to how they utilize metrics, but it could also be due to how heavily quotes are relied upon in this category; a case study of a Politico article demonstrated why this is relevant.

Politico Case Study: Why Upworthy hides its traffic stats from staffers

Politico's 2014 article starts by introducing their subject:

> "It turns out data isn't always useful, at least according to Upworthy director of business intelligence Daniel Mintz."

This begins a trend within the article that typifies the targeted documents. When another outlet's implementation and use of metrics is mentioned, it is never evaluative, and if there is a judgment, it is never in the authors words. This holds true for opinion, column, and blogs as well. Often, targeted documents exist solely to relay interesting case studies of how media entities use metrics:

> ""People love to obssess over 475,000 [page views] versus 375,000," he said. "That's not very useful. We went through a very long process of hiding numbers from our curators where we replace them with words. We haggled over the terminology for several months and ended up with things like 'bestish' because we didn't want to communicate more certainty than we as analysts really had... 'probably worse,' 'maybe worse,' 'definitely worse.' Those are literally the words that curators see. Then our product folks realized that some visual cues would help with that and then they asked me for pictures of me...so now on the page they have pictures of me with thumbs up and thumbs down signs."'

Aside from the cartoonish visuals, this language constitutes the bulk of this category: mechanical, descriptive and in-the-now. Contrast this with the charged, future-minded write ups found in the print and digital categories.

> "Picking the metrics that align with his site's goal is critical, Mintz said in explaining Upworthy's recent shift in focus to tracking readers' attention minutes as opposed to clicks-per-article."

[11] https://lukedanielcross.github.io/files/targetedlda.html

Untargeted LDA[12]

Topic 3 stands out in this category for being the closest connection between the outlet(newsroom, journalist, story) and the public (public, audience, people) that has been demonstrated thus far. Both clusters of terms are highly salient within that topic. Furthermore, Topic 3 contains terms possibly related to experimentation and improvement (project, organization, study, question). A column from Observer illustrates this trend.

Observer Case Study: What to Measure? The Question Journalism Has to Answer

Untargeted documents are, as one would imagine, more oriented towards broad stroke critiques. An interesting trend, however, is a seeming acceptance of metrics and the desire to refine them:

> "When I see Marc Andreeseen's intelligent critiques of journalism, particularly the bad metrics of current journalism, l agree. But the other side of me thinks: well, what the hell else are people supposed to do? What am I—or any publisher—supposed to measure then? How should media outlets in 2014 determine what "success" looks like and create employee compensation parameters around it? If pageviews or uniques aren't it, what should it be? Gut instincts? Length? Comments? What?

Most documents don't propose solutions, but it is worth noting that untargeted documents rarely propose a question that needs solving. Unlike most other categories, it seems untargeted documents rarely speak to concerns of a dying industry or economic challenges. Instead, they typically fall back on an analysis of their industry:

> "It's the lack of a clear, easy answer to these questions that has us in the state we are currently in. As much as I've criticized it, I empathize with the quandary. Page views are easy to track. Quality is harder. It's not only relative and subjective, it takes time and resources that many people don't have. It's easier to fall back on "whatever everyone else is doing.""

Reflective LDA[13]

Small sample size renders LDA null.

NYT Case Study: Metrics Are Everywhere in Media. Here's How They Help.

In an article for the Technology section of NYT's website, Newsroom and Product Analytics Manager Kathy Zhang discussed the metrics software their team uses and how it integrates with newsroom processes. Within the tech-oriented discussion of Zhang's position and equipment, possible concerns of the influence of metrics are occasionally hinted at; Zhang almost immediately follows these concerns with explanations:

> "We don't let pageviews dictate our coverage or approach, but we do use data to understand reader engagement and the reach of our stories ... But analytics allow us to serve our audience better ... Though many lament that access to analytics can make

[12]https://lukedanielcross.github.io/files/untargetedlda.html
[13]https://lukedanielcross.github.io/files/introspectivelda.html

> a publication hyper-focused on pageview volume to the detriment of reader engagement, at The Times we're cognizant of those pitfalls, and we structure our analyses to avoid them. The close partnership our newsroom editors have with the analytics team makes both our editors and our analysts think smarter about what our data tells us and how we can best deliver our journalism."

While Zhang openly addresses common fears and concerns, these mentions are paired with an immediate rebuttal. Any discussion of metrics in a non-technical sense it mitigated with en explanation, assuaging concerns of how metrics aren't infecting the journalistic process while also tacitly asserting that metrics are de facto harmful for reporting. This is further reinforced by Zhang's appraisal of who uses metrics:

> "Editors are some of our best analytics consumers. They constantly monitor our dashboards to make real-time judgments about how best to engage our readers. News desk editors modify headlines and packaging to draw attention to stories. Search specialists take advantage of trending search terms to expose our stories to a wider audience. Social media editors follow the conversation and find the best moments to showcase our journalism. Analytics are crucial to the full life cycle of our coverage."

The detailed list provided above appears rich and diverse in titles, but note that no mention of reporters are made — only editors and more technical positions. As an introspective article, any statements that divert from technocratic descriptions are defanged by immediately addressing potential concerns.

Hypotheses and the Data

H1: Print outlets will publish less about metrics

Corpus distribution supports H1, though the rationale and influences driving this trend is unknown.

H2: Financial topics will be a small minority of the text, except for business and strategy documents

Explicit mention of financial topics exist in print as well as business and strategy, though it could be argued that struggles to finance media are implicit in the complaints found in the digital outlets.

H3: Targeted and reflective documents will be descriptive, untargeted will be prescriptive

Excluding reflective documents for aforementioned size issues, targeted documents were highly descriptive, but targeted documents rarely pushed the bounds of prescriptive language. Problems were often identified, which is a normative step beyond what was exhibited in targeted documents, but full fledged calls for change were rare.

Limitations and Future Work

As noted in the Corpus Distribution, the websites of print newspapers are not carbon copies of the stories printed in their physical papers. There is the chance that a physically printed article does not make it to digital publication or vice-versa, possible skewing the results. This can be somewhat accounted for with a large enough sample, but the only holistic means of addressing

this blindspot would be collecting data from archived copies of the papers themselves. Provided digital archives of old editions exist for the relevant papers, this could be accomplished in the future using optical character recognition software.

Given the exploratory nature of this work, and subsequently the small sample size employed, expanding the webcrawler and LDA modeling method to include more outlets is the most obvious expansion. Leveraging outlets from other English speaking countries, with the proper controls and data stratification, could be an effective means of accomplishing this.

LDA, and topic modeling as a field, is still nascent and expanding. Many submodels and implementations of LDA exist, and, given the sometimes inscrutable nature of text-to-data analysis, best practices are still unclear for many of these implementations[14] Testing the established and future corpora for this work has the possibility of uncovering semantic connections missed in this implementation.

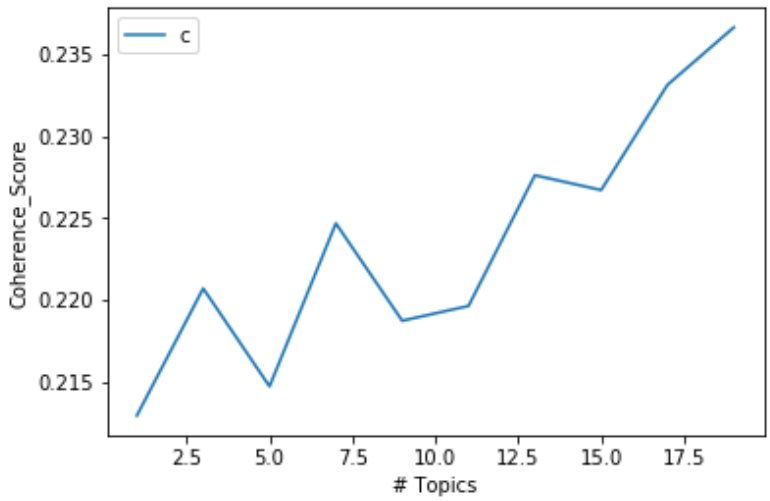
Lastly, supplementing the theoretical and exploratory findings of these models with more concrete understandings of how and why the relevant documents were published, such as focused qualitative interviews with the authors or outlets, could further elucidate the findings of these models.

[14]Mimno, D., & Mimno, M. D. (2013). Package 'mallet'. A wrapper around the Java machine learning tool MALLET ; Biggers, L. R., Bocovich, C., Capshaw, R., Eddy, B. P., Etzkorn, L. H., & Kraft, N. A. (2014). Configuring latent dirichlet allocation based feature location. Empirical Software Engineering, 19(3), 465-500.

Appendix A: Coherence Scores and K-Topics Charts

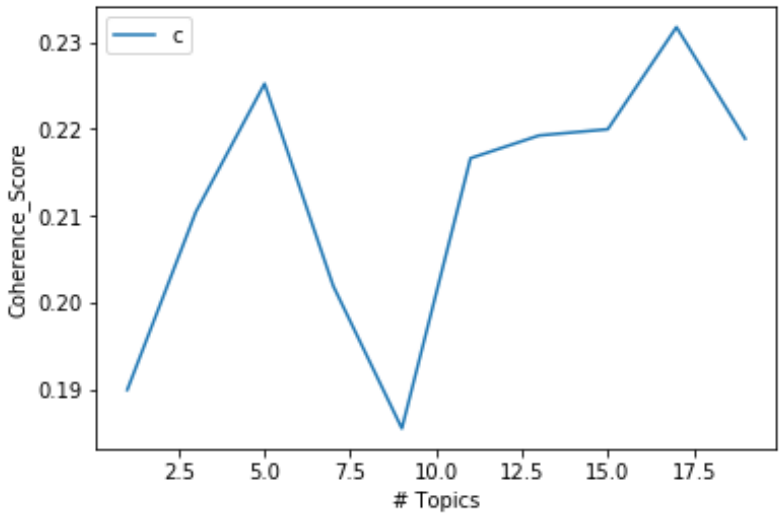
Digital Model

```
Perplexity:  -7.593107979077423

Coherence Score:  0.3331130089253538
```



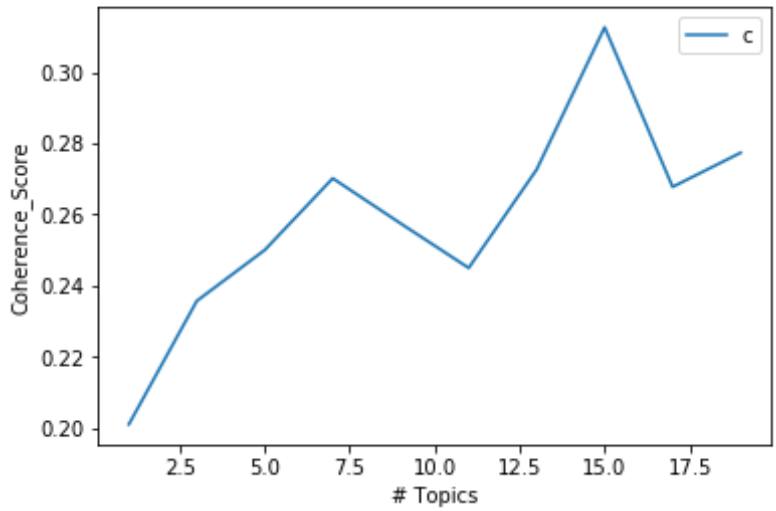
Print Model

```
Perplexity:  -7.281136107458294

Coherence Score:  0.28725388216399544
```



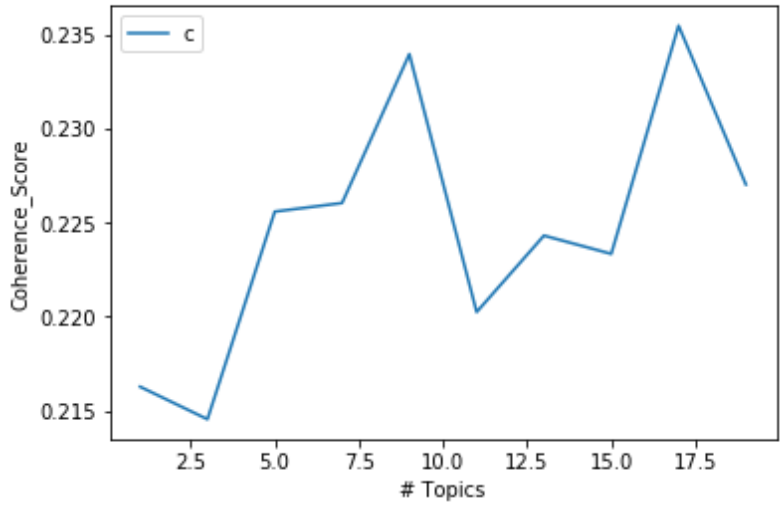
Business and Strategy Model

```
Perplexity:  -6.856167276615413

Coherence Score:  0.4470776887734184
```



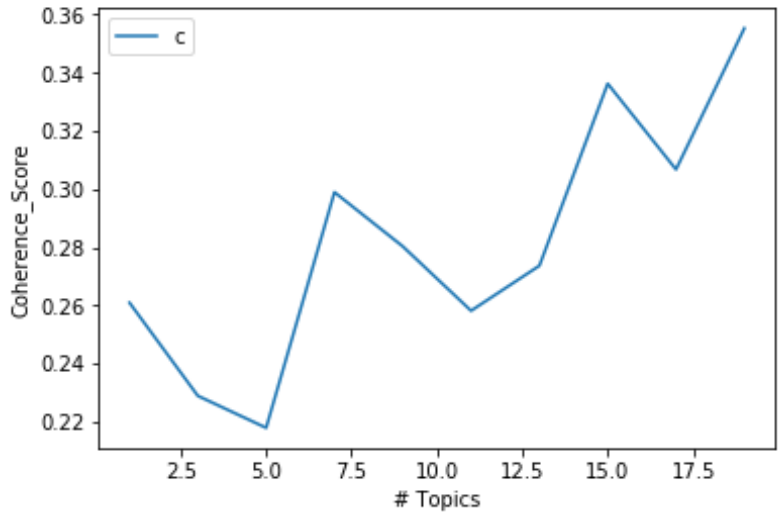
Targeted Model

```
Perplexity:  -7.430243918064692

Coherence Score:  0.3003813848624664
```



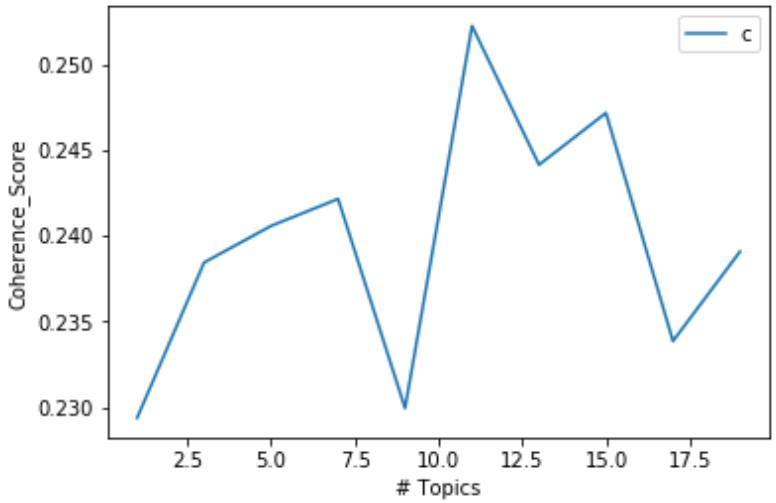
Reflective Model

```
Perplexity:  -6.6946870215704015

Coherence Score:  0.369470650494871
```



Untargeted Model

```
Perplexity:  -7.558273607405024

Coherence Score:  0.2923102521137988
```



Appendix B: Outlets and Number of Docs

```
##                          Outlet Number_of_Docs
## 1                Washington Post              3
## 2                       Buzzfeed              2
## 3                Huffington Post              5
## 4                            Vox              2
## 5                Chicago Tribune              1
## 6                   The Guardian              4
## 7                  Seattle Times              3
## 8                       Newsweek              2
## 9                    Denver Post              0
## 10                  Mercury News              0
## 11              Washington Times              0
## 12                  Miami Herald              0
## 13                 The Intercept              0
## 14             Chicago Sun Times              0
## 15                       Observer              2
## 16                  Daily Herald              0
## 17                       Mashable              2
## 18                            MIC              0
## 19                  Boston Globe              0
## 20                       Atlantic              1
## 21                       LA Times              0
## 22                        Newsday              0
## 23                        Complex              0
## 24                   Daily Beast              0
## 25                        TheHill              0
## 26                     refinery29              2
## 27                Tampa Bay Times              9
## 28             Houston Chronicle              0
## 29                      AZ Cental              0
## 30                          Slate              0
## 31                           Vice              1
## 32                          salon              0
## 33                         Gawker              4
## 34                     TechCrunch              2
## 35                       UpWorthy              4
## 36                       Politico              0
## 37                         Forbes              0
## 38                       Economist              1
## 39               Pacific Standard              3
## 40                    ArsTechnica              1
## 41 Atlanta Journal Constitution              0
## 42                  Star Tribune              0
## 43                       Inquirer              1
## 44                  Plains Dealer              0
## 45          Morning News (Dallas)              0
```

```
## 46         Oregonian (Portland)             0
## 47       Orange County Register             0
## 48    Post-Dispatch (St. Louis)             0
## 49     Bee (Sacramento, Calif.)             0
## 50      Star (Kansas City, Mo.)             0
## 51 Journal Sentinel (Milwaukee)             0
## 52                 Miami Herald             0
## 53                   The Nation             0
## 54            RealClearPolitics             0
## 55    Christian Science Monitor             0
## 56                    Breitbart             0
## 57                  Epoch Times             0
## 58        American Conservative             0
## 59                     the Blaze             0
## 60               The Federalist             0
## 61                       uproXX             0
## 62                    The Verge             0
## 63                      gizmodo             0
## 64                        Axios             0
## 65               The New Yorker             2
## 66               New York Times             5
```